

Bahram Behzadian

Reinforcement Learning · Post-Training · Model-Based RL

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Research Summary

I work on reinforcement learning and sequential decision-making. The through-line across my research is policy improvement when the model of the world or the evaluator is imperfect: robust MDPs when the dynamics are uncertain, and truncated or approximate evaluators when the reward signal is unreliable. My recent work studies policy improvement under verifier-style rewards, connecting to post-training and RL for reasoning models. Earlier work spans robust MDPs, learning under uncertainty, and partial observability. I am increasingly focused on model-based RL and reasoning systems that plan under imperfect models.

Publications

Bahram Behzadian, Brett Daley, Gopeshh Subbaraj, Houssam Nassif. UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards. *ICLR 2026 RSI Workshop; NeurIPS 2026 (under review)*.

Decomposes finite-depth evaluator error into a Bellman residual and a truncation bias, propagated through a conservative policy improvement bound. Connects to RLVR and verifier design.

Bahram Behzadian, Marek Petrik, Chin Pang Ho. Fast Algorithms for L-infinity-Constrained S-Rectangular Robust MDPs. *Neural Information Processing Systems (NeurIPS)*, 2021.

Scalable algorithms for robust policy computation under structured uncertainty.

Bahram Behzadian, Reazul Hasan Russel, Marek Petrik, Chin Pang Ho. Optimizing Percentile Criterion using Robust MDPs. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021.

Robust decision-making under risk-sensitive and quantile-based objectives.

Bahram Behzadian, Soheil Gharatappeh, Marek Petrik. Fast Feature Selection for Linear Value Function Approximation. *International Conference on Automated Planning and Scheduling (ICAPS)*, 2019.

Representation and feature selection for value-based reinforcement learning.

Bahram Behzadian, Pratik Agarwal, Wolfram Burgard, Gian Diego Tipaldi. Monte Carlo Localization in Hand-Drawn Maps. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2015.

State estimation under partial observability with noisy map representations.

Reazul Hasan Russel, Bahram Behzadian, Marek Petrik. Optimizing Norm-Bounded Weighted Ambiguity Sets for Robust MDPs. *NeurIPS Workshop on Safety and Robustness in Decision-Making*, 2019.

Bahram Behzadian, Marek Petrik. Feature Selection by Singular Value Decomposition for Reinforcement Learning. *ICML Workshop on Prediction and Generative Modeling*, 2018.

Bahram Behzadian, Marek Petrik. Low-Rank Feature Selection for Reinforcement Learning. *ISAIM*, 2018.

Federico Boniardi, Bahram Behzadian, Wolfram Burgard, Gian Diego Tipaldi. Robot Navigation in Hand-Drawn Sketched Maps. *European Conference on Mobile Robotics (ECMR)*, 2015.

Experience

2022–Present **Research Scientist**, *Meta*, Menlo Park, CA

- Build large-scale experimentation and evaluation infrastructure: metric correctness pipelines, anomaly and data-quality monitoring, and ML auditing for production systems.
- Design statistically grounded evaluation frameworks that surface when large-scale learning systems are unreliable, biased, or being gamed.

2018 **AI Engineer Intern**, *Envio, Inc.*, Dover, NH

- Learning-based methods for vehicle routing and scheduling as constrained sequential decision-making problems.
- Combinatorial planning and long-horizon trade-offs under execution uncertainty.

Education

2015–2022 **Ph.D. in Computer Science**, *University of New Hampshire*

Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik). M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.

2010–2013 **M.S. in Machine Automation**, *Tampere University*

Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire

Thesis and Dissertation Fellowship, Tampere University of Technology

Skills

Reinforcement Learning: robust MDPs, learning under uncertainty, partial observability, model-based RL, planning, policy improvement under imperfect evaluators, verifier-style rewards and RLVR.

Post-training: DPO, PPO, GRPO, reward modeling, evaluation methodology.

Engineering: Python, PyTorch, large-scale experimentation and evaluation infrastructure.